BackupAndRestoreScripts

Scripts for backing up, moving and restoring software on Beckhoff TwinCAT 3 IPCs. Each tool has its own folder, and each folder's README covers setup and details.

Folder	What it's for
`BootFolderAndTF2000` (Windows, Linux)	Copies the TwinCAT Boot folder (PLC + I/O config) and the TF2000 HMI Server data from a bench system to machine IPCs on a USB stick, with automatic backup and rollback.

BootFolderAndTF2000 in short

You build the software on a **bench system**, then put it on each **machine IPC** with a USB stick. No engineering laptop or TwinCAT XAE is needed at the machine.

1. **On the bench:** plug in a USB stick and run `Create-UpdatePackage.bat`. It briefly stops TwinCAT and the HMI Server and copies the Boot and HMI folders. It also saves a checksum for every file and records the TwinCAT / TF2000 versions, then starts TwinCAT and the HMI Server again and zips everything into a single file, `<USB>:\TwinCAT_Update.zip`.
2. **At the machine:** plug in the stick and press the update button on the HMI. The PLC runs `Apply-Update.bat`, which:
   - unzips the package to a local folder, checks it is complete and that the TwinCAT / TF2000 versions match this machine (if anything is wrong, it stops and changes nothing)
   - stops TwinCAT and the HMI Server, backs up the current version, and copies in the new one
   - checks every file, restores the backup automatically if anything fails, then switches TwinCAT back to Run mode and starts the HMI Server again (Windows; the Linux version restarts the controller)
3. **Check the result** in `C:\TwinCAT_UpdateBackups\last_update_status.txt` on the machine, or in the log copied back to the stick under `TwinCAT_Update_Logs\`.

Before the first use, check the paths in `BootFolderAndTF2000\Windows\Config.ps1`. Then copy the folder to a fixed local path on each machine (e.g. `C:\TwinCAT_UpdateTool\`).
BootFolderAndTF2000/Windows/README.md covers the full setup, the PLC trigger code, optional arguments, logs and manual rollback.

Optional on Windows: `BootFolderAndTF2000\Windows\UpdateScreen\` shows a full-screen "system is being updated, do not power off" page with live progress on the machine's HMI while the update runs. The update itself does not depend on it. See the Windows README.

The Linux version works the same way with `Create-UpdatePackage.sh` / `Apply-Update.sh` and `Config.sh`, using a `.tar.gz` archive instead of a zip. The result file is `/var/lib/TwinCAT_UpdateBackups/last_update_status.txt`.

Requirements

- Windows IPCs with TwinCAT 3 installed. The scripts use the ADS DLL that comes with TwinCAT, so you don't need to install anything extra.
- Windows PowerShell 3.0 or newer (built into Windows). The scripts need administrator rights. When the PLC starts them, they run as SYSTEM.
- Linux IPCs (Beckhoff RT Linux / Debian with systemd): bash, `rsync` and an ADS command-line tool such as `adstool`. See the Linux README.
- The bench system and the machines should be the same IPC model, with the same TwinCAT and TF2000 versions.